ProtoCBM: Improving Localization of Concept Bottleneck Models with Prototype Learning

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Abstract

Concept Bottleneck Models (CBMs) improve the interpretability of deep neural networks in image classification by introducing an intermediate layer of human-understandable concepts, before making a final prediction from these concepts. While allowing for easier post-hoc analysis, CBMs often fail to spatially ground their concept predictions in meaningful image regions, suggesting reliance on spurious correlations. To address this issue, we introduce ProtoCBM, a novel extension of CBMs. Our model uses prototype learning for concept prediction, representing each concept with a learnable prototype vector. Using the same training signals as vanilla CBMs, the prototypes are optimized to activate in image patches semantically related to the concept, thus explicitly localizing concepts. We further extend this framework with LocoCBM, which imposes explicit supervision on concept locations. Additionally, we introduce two metrics to evaluate concept localization. Experiments on the CUB, SUB and Waterbirds datasets demonstrate that our approach outperforms CBMs in localization, while offering competitive class and concept accuracy. Moreover, it exhibits an improved robustness to spurious correlations and distribution shifts via concept substitutions. Our results show that prototype-based concept localization is a promising direction to improve the interpretability and faithfulness of CBMs.

1. Introduction

Despite their astounding performance on visual recognition tasks like image classification, deep neural networks are often considered black-box models [5], as their high-dimensional and non-linear representations inhibit the interpretability of their predictions. This makes it substantially harder to analyze faulty predictions of models. Concept Bottleneck Models (CBMs) [11] have emerged as a promising approach to address this issue by introducing an intermediate concept layer between input data and output predictions. By predicting human understandable concepts as a bottleneck layer, they increase transparency, albeit with reduced task performance compared to their black-box counterpart [9, 12]. This intermediate layer enables post-hoc analysis and human intervention by either interpreting or modifying intermediate predictions [11]. However, CBMs do not guarantee spatial grounding of concepts, resulting in wide-spread attention over irrelevant image regions when predicting concepts [13]. This can also be seen in Fig. 1, displaying the poor localization capacity of vanilla CBMs. Recent works [13, 18, 19] have shown CBMs to rely on spurious correlations, rather than semantically meaningful image regions. As a result, the predicted concepts often lack visual grounding, reducing the interpretability and faithfulness of models.

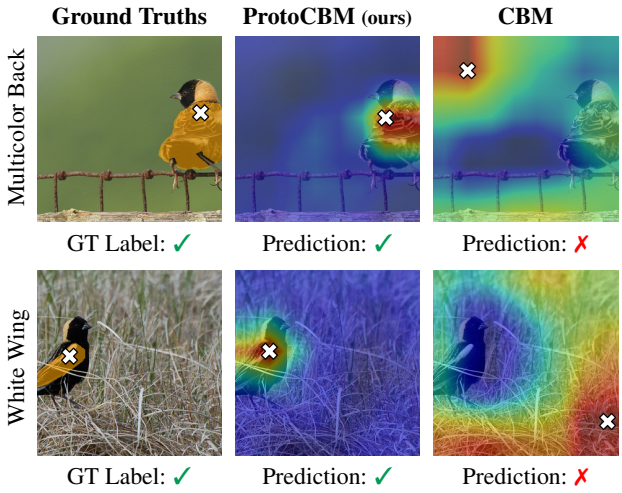


Figure 1. **ProtoCBM concept maps.** Activation maps of our approach during concept prediction (red indicates higher activation) and the respective predictions of concept presence (✓) or absence (✗). ProtoCBM correctly localizes and spatially grounds its predictions. Conversely, CBM [11] fails to correctly predict concept presence, as it attends to irrelevant image regions.

Based on these findings, we hypothesize that explicit concept localization will amend these shortcomings and result in higher robustness of the model, relying less on spurious correlations. In this work, we propose ProtoCBM, a novel extension of CBMs that encourages locality respecting concept predictions through prototype learning [24, 27],

requiring no additional supervision signal beyond the standard CBM framework. ProtoCBM represents each concept using a learnable prototype vector that is trained to activate on concept-related regions of an image. Each prototype is convolved with the image feature map to obtain patch-wise activations, which are pooled into one concept score. When substituting the concept mapping of CBMs with our approach, it allows for concept predictions while producing localization maps that are more grounded in the correct image patches. These intermediate concepts are subsequently used for class prediction using a fully connected layer [11], as done in vanilla CBMs. In addition, we introduce two complementary metrics that quantify distance-based and area-based (mIoU [17]) concept localization performance. Finally, we introduce explicit localization supervision as a further extension to ProtoCBM, evaluating the effect of guided localization on class and concept accuracy.

We train our model in several settings, following the training procedures from Koh et al. [11], using the CUB [26], SUB [2] and Waterbirds [21] datasets as benchmarks. Our results demonstrate competitive class and concept accuracy and a clear improvement in concept localization on the CUB benchmark. Our model improves concept grounding by up to 28.3 pixels in localization distance and 0.8 percentage points in mIoU scores. These metrics are further supported qualitatively by more precise concept activation maps, as seen in Fig. 1. Additional findings on the SUB and Waterbirds datasets show improved sensitivity to artificial concept substitutions, as well as greater robustness to spurious correlations between concepts.

The key contributions of this work are as follows: *(1)* We propose ProtoCBM, a variation of CBMs that utilizes prototype learning for concept mapping. *(2)* We extend this framework with LocoCBM, which imposes explicit supervision on concept locations. *(3)* Additionally, we introduce two metrics that quantify distance-based and area-based concept localization. *(4)* We conduct extensive evaluations on the CUB, SUB and Waterbirds benchmarks, where we demonstrate the effectiveness of our approach w.r.t. task and concept accuracy, concept localization, and robustness.

2. Related Work

Concept Bottleneck Models (CBMs). Koh et al. [11] introduced CBMs to improve interpretability in neural networks. The model is forced to predict concepts understandable by humans, before making a final, task-specific prediction. To do so, CBMs are effectively split into two parts, a concept mapper and a class predictor. The former maps images to concept scores, while the latter takes these intermediate concept scores as input and outputs a class label prediction. This creates a bottleneck layer in the concept space of images, allowing for post-hoc inspection and intervention.

Standard CBMs are subjected to multiple drawbacks. Learning on class or image-specific concepts requires significantly more annotations than plain class labels. While good concept annotations may reduce the amount of required data, this still adds significant annotation overload [11]. Moreover, CBMs typically reduce the task accuracy [9, 12], which decreases their applicability in real-world scenarios. Approaches like [14, 29, 30] remedy this via frameworks for transforming general neural networks into CBMs with equal task performance, utilizing concept embeddings and transfer learning. This amends the need for concept-level annotations and renders the bottleneck models more scalable to real-world tasks by reducing reliance on concept-annotated datasets. Current research on CBMs focuses on mitigating these performance drops and label dependencies via hybrid concept banks [9] or semi-supervised concept learning [12], as well as improving the post-hoc concept intervention process [16] through deferred learning on human accuracy.

In contrast to these recent extensions, we rely on qualitative, supervised concept annotations and utilize the standard pipeline of Koh et al. [11]. We primarily focus on the mapping of inputs to the concept space, and investigate the locality of attention in this part of CBMs.

Interpretability of Concept Bottleneck Models. CBMs offer enhanced interpretability through the bottleneck layer, in which human understandable concepts are represented. This allows post-hoc cause analysis of failed classifications. However, Raman et al. [18, 19] argue that these concepts do not faithfully capture the internal behavior of the model. Their work focuses on analyzing whether the concept mapping utilizes concept-relevant features in the image. To this end, they mask out concepts or perturb concepts in nearby regions. They find that CBMs often fail in these settings, concluding that these models use spurious feature correlations to predict concepts, invalidating their interpretability. Other works further investigate information leakage between the concept mapper and class predictor [8], showing that the robustness of CBMs remains in doubt. Our experiments thus investigate potential spurious connections on appropriate benchmarks [21] to further explore this topic in our setting.

Aside from score-based analysis of concepts, another widely used interpretability method is the visualization of concept attendance. Margeloiu et al. [13] utilize gradient-based methods to visualize the concept-wise attention of the model in an image. While gradient methods such as Grad-CAM [22] are not equally expressive as attention maps, the findings of [13] show that CBMs rarely focus on the actual part of the image depicting the concept. Instead, CBMs rather attend to various unrelated regions of the image (*cf*. Fig. 1). This suggests reliance on spurious correlation [19], rendering concepts less interpretable. Thus, investigation of correct concept localization is central to our analysis.

Prototype Learning. Prototypical networks [24] learn prototype representations of data, which are subsequently used

for classification by computing the distances in the learned metric space between inputs and individual prototypes. This approach is commonly adapted in research with limited data or need for faithful explanations, due to increased robustness for open-set recognition and few-shot learning [7, 23]. Learned prototypes can be treated as class centroids [1] or as piece-wise representations that describe different parts of an object [4]. The latter work simulates a natural classification process by dissecting an object into different parts that are associated with descriptive prototype vectors. The network then aggregates the part-wise information to categorize the input into one of multiple classes. Similarly, Xu et al. [27, 28] associate their prototype vectors with spatial features of image patches. Each vector is chosen to represent a distinct concept and is trained to have a high activation when encountering a corresponding feature patch. This allows for computation of similarity maps between concepts and input images by convolving the prototypes with the input images. The authors pair this local prototype training with a global classification approach that encodes the input image into feature vectors to predict the classes in a zero-shot setting.

Prototype dissection of classification problems into concepts allows a visual reasoning process and produces intermediate results usable for final classification. Moreover, it produces attention maps for each concept prototype, providing enhanced interpretability through visual inspection of the model's localization. However, this paradigm remains largely unexplored within CBMs. Our approach fills this gap by utilizing prototype learning as a concept mapper network, while discarding the global branch of the architecture.

3. Method

We now introduce all components of our method in detail.

3.1. Background

Concept Bottleneck Models (CBMs). The central idea behind CBMs [11] is to introduce an intermediate concept layer during the classification of images of size $H \times W$. Specifically, instead of learning a function $f : \mathbb{R}^{H\times W} \to \mathbb{R}$, we learn a mapping from inputs to C concepts consisting of a backbone $b : \mathbb{R}^{H\times W} \to \mathbb{R}^{F\times H'\times W'}$ and concept mapper $m : \mathbb{R}^{F\times H'\times W'} \to \mathbb{R}^{C}$. Furthermore, we train a classifier $c : \mathbb{R}^{C} \to \mathbb{R}$, such that $f = c \circ m \circ b$. Intuitively, the enforced reduction to a lower-dimensional concept representation ensures that the classification decision is only based on the specified concepts, not spurious correlations.

Prototype Learning. Xu et al. [27] introduce ProtoMod for localized feature detection in their Attribute Prototype Network (APN). Given the output of a CNN backbone z of size $F \times H' \times W'$, they propose a learnable prototype vector $p_i \in \mathbb{R}^F$ for each concept $i = 1, \dots, C$. Here, F denotes the feature dimensionality; H' and W' refer to the feature map height and width, respectively. For a given concept i, we calculate the similarity A_i of its prototype vector to each location $u \in \Omega'$ in the feature map z with $\Omega' = \{1, \dots, H'\} \times \{1, \dots, W'\}$. This corresponds to a 1×1 convolution with the kernel p_i

$$A_i(z_u) = \langle p_i, z_u \rangle. \tag{1}$$

We subsequently apply max-pooling to find the maximum activation

$$a_i^+(z) = \max_{u\in\Omega'} A_i(z_u) \tag{2}$$

for each concept. The final concept score is obtained by applying a sigmoid function σ, *i.e.* $a_i(z) = \sigma(a_i^+(z))$.

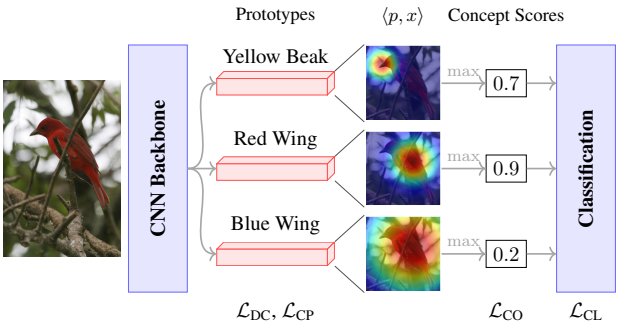


Figure 2. **ProtoCBM architecture.** The input image is passed through the backbone b to obtain a feature map over which we convolve concept-specific prototype vectors, yielding activation maps. Subsequently, max-pooling is applied to obtain concept scores, which are finally fed into a classifier to produce class predictions.

3.2. ProtoCBM (Ours)

We introduce ProtoCBM, combining CBMs with prototype learning. For that purpose, we exchange the feed-forward layers used as the concept mapper m with the prototype layer described in Section 3.1. Both the backbone b and classifier c are identical to vanilla CBMs. The entire ProtoCBM architecture is visualized in Fig. 2.

Loss. To not only predict the correct class label but also produce valid intermediate concepts, we apply two binary cross entropy loss terms

$$\mathcal{L}_{\text{BCE}}(y, \hat{y}) = -\left[y \log(1 - \hat{y}) + (1 - y) \log \hat{y}\right] \tag{3}$$

with labels $y \in \{0, 1\}$ and predictions $\hat{y} \in (0, 1)$. Concretely, for concept labels y_{CO}, class labels y_{CL}, and the model's concept predictions $\hat{y}_{\text{CO}} = m \circ b(x)$ we compute the classification loss

$$\mathcal{L}_{\text{CL}}(y_{\text{CL}}, \hat{y}_{\text{CO}}) = \mathcal{L}_{\text{BCE}}(y_{\text{CL}}, c(\hat{y}_{\text{CO}})) \tag{4}$$

and the concept loss

$$\mathcal{L}_{\text{CO}}(y_{\text{CO}}, x) = \mathcal{L}_{\text{BCE}}(y_{\text{CO}}, m \circ b(x)). \tag{5}$$

For additional regularization of the learned prototypes, we implement the corresponding loss terms proposed by [27]. First, they apply an attribute decorrelation criterion $\mathcal{L}_{\text{DC}}$ based on [10]. This is added to avoid the prototypes picking up on correlation with other groups (*i.e.* a blue head might indicate a blue body and therefore both prototypes might attend to the bird's head). Considering there are multiple attributes concerned with an individual body part, we introduce groups of attributes $G_1, \dots, G_L$ that correspond to L different bird parts (see Appendix F for details). To avoid decorrelation within a group, we use these groups to decorrelate between them. Concretely, given prototype vectors $\{p_{l:i'} \mid i' = 1, \dots, |G_l|\}$ for group l concatenated into P_{G_l}, the attribute decorrelation loss can be formulated as

$$\mathcal{L}_{\text{DC}} = \sum_{j=1}^{F} \sum_{l=1}^{L} \|P_{G_l}[j]\|_2 \tag{6}$$

where $P_{G_l}[j]$ is the j-th feature of the prototype vectors in group l. This criterion penalizes occurrences of a feature across groups, while multiple occurrences in the same group lead to a smaller penalty. Finally, we apply compactness regularization [27]. Given the activation maps $A_i(z)$ for prototype p_i, we penalize scores depending on the squared distance to the maximum activation $a_i^+(z)$ according to

$$\mathcal{L}_{\text{CP}}(x) = \sum_{j=1}^{F} \sum_{u' \in \Omega'} A_i(b(x)_{u'}) \|u_i - u'\|_2^2 \tag{7}$$

given the location of the highest activation $u_i = \text{argmax}_{u \in \Omega'} A_i(b(x)_u)$. Note that these loss terms do not require additional supervision compared to vanilla CBMs [11]. Considering the modular approach of CBMs, there are multiple options to combine the presented losses. For this purpose we summarize the losses imposed on concepts as

$$\mathcal{L}_{\text{C}}(y_{\text{CO}}, x) = \lambda_{\text{CO}} \mathcal{L}_{\text{CO}}(y_{\text{CO}}, x) + \lambda_{\text{DC}} \mathcal{L}_{\text{DC}} + \lambda_{\text{CP}} \mathcal{L}_{\text{CP}}(x) \tag{8}$$

with weighting parameters λ_{CO}, λ_{DC}, λ_{CP}. We adapt the training procedures proposed by Koh et al. [11] for a given class label y, which we enumerate in the following:

Joint. The entire network is trained concurrently and we apply a combination of all loss terms:

$$\mathcal{L}_{\text{Joint}}(x, y_{\text{CO}}, y_{\text{CL}}) = \mathcal{L}_{\text{CL}}(y_{\text{CL}}, m \circ b(x)) + \mathcal{L}_{\text{C}}(x, y_{\text{CO}}) \tag{9}$$

Sequential. Sequential training consists of two phases: First, the concept mapper is trained using $\mathcal{L}_{\text{C}}(y_{\text{CO}}, x)$, before it is frozen and the classifier is optimized by $\mathcal{L}_{\text{CL}}(y_{\text{CL}}, m \circ b(x))$.

Independent. Similar to sequential, independent training is split into two phases. During inference, we also use the predicted concepts for classification, however we train on the ground-truth concepts y_{CO}, *i.e.* $\mathcal{L}_{\text{CL}}(y_{\text{CL}}, y_{\text{CO}})$.

3.3. LocoCBM (Ours)

To strictly enforce localized concept detection, we additionally introduce LocoCBM, which imposes supervision on part locations using ground-truth locations. Specifically, we utilize the key point annotations g_l (*cf*. Appx. F) for each group l by scaling them to their location in the $H' \times W'$ feature map. We then create a gaussian heatmap A_l^{GT} centered around this point with values

$$A_l^{\text{GT}}(u) = \exp\left(-\frac{\left\|u - g_{l(i)}\right\|_2^2}{2\sigma^2}\right) \tag{10}$$

at each point $u \in \Omega$ with $\sigma = 1$. We normalize the gaussian heatmaps so that $\sum_{u' \in \Omega} A_{l(i)}^{\text{GT}}(u') = 1$, where $l(i)$ denotes the group of which i is a member. To enable comparison, we also normalize our model's heatmap A using Softmax

$$A_i^x(u) = \frac{\exp A_i(x_u)}{\sum_{u' \in \Omega} \exp A_i(x_{u'})}. \tag{11}$$

Since both $A_{l(i)}^{\text{GT}}$ and A_i^x are now probability distributions over $u \in \Omega$ for each concept i, we can formulate our distance criterion as the KL-divergence between them

$$\mathcal{L}_{\text{DL}}(x, g) = \frac{1}{C} \sum_{i=1}^{C} \mathcal{D}_{\text{KL}}\left(A_i^x \| A_{l(i)}^{\text{GT}}\right). \tag{12}$$

This loss term is added to our concept losses $\mathcal{L}_{\text{C}}$ with the corresponding weight λ_{DL}.

4. Experiments

In this section, we explain our experimental setup and present the results of our method. We compare our approach in all settings to Concept Bottleneck Models (CBMs) [11].

4.1. Experimental Setup

Datasets. We conduct experiments on multiple datasets to assess the effectiveness of our approach in concept-based image classification. ProtoCBM and LocoCBM are trained on the Caltech-UCSD Birds-200-2011 (**CUB**) dataset [26], comprising 11 788 bird images of 200 bird classes annotated with 112 binary bird concepts, grouped into 13 concept parts. To mitigate annotation noise, we perform intra-class majority voting of concepts, ensuring that birds of the same class are assigned to the same concept labels, following [11]. To investigate concept localization, we utilize the part annotations of [26], where each concept part is assigned to one pixel location within the image. We further use **CUB-70** [3], a subset of CUB containing segmentation masks of 11 concept parts for 70 bird classes, to assess area-based localization. We refer to the supplementary (*cf*. Appx. F) for further details. The robustness of our approach to distribution shifts

via concept substitutions is evaluated on the **SUB** dataset [2], which contains synthetic bird images where a target concept is substituted with another value atypical for its bird species. Finally, we utilize **Waterbirds** [21], an adaption of CUB in which backgrounds are replaced with land and water scenes of Places [32] to assess the robustness to spurious correlations between bird species and backgrounds.

Metrics. Following prior work [11, 27], we employ class-wise ($\mathcal{A}_{\text{cls}}$) and concept-wise ($\mathcal{A}_{\text{con}}$) accuracy between model predictions and ground-truth labels to quantify performance in concept-based image classification. For $\mathcal{A}_{\text{con}}$, intermediate concept logits are passed through a sigmoid function and thresholded at 0.5 to obtain binary predictions, which are subsequently compared to the ground-truth concept labels. While accurate localization of concepts is vital to faithful concept-based models, systematic evaluation metrics of concept localization are underexplored [15]. [27, 31] propose to assess the Intersection over Union (IoU) [17] between bounding boxes drawn around the part location annotations of CUB and the area of highest concept activation, however, concerns arise regarding its reliability (*cf*. Appx. E). Therefore, we propose two complementary metrics to evaluate concept localization based on concept activation heatmaps.

First, we assess *distance-based localization*, which measures the Euclidean distance between ground-truth part annotations and locations of maximal activation in concept heatmaps, averaged over all parts. Let $\mathcal{H}_l = \{H_{l:1}, \dots, H_{l:n_l}\}$ denote the set of n_l concept activation heatmaps belonging to concept group G_l, stemming from either CBM $\left(H^C\right)$ or our model $\left(H^P\right)$, *cf*. Appx. Sec. D. Within it, we select the index o_l of the heatmap $H_{l:o_l} \in \mathbb{R}^{H \times W}$ with the maximum concept score $a_{l:o_l}$

$$o_l = \underset{i' \in \{1,\dots,n_l\}}{\operatorname{argmax}} \; a_{l:i'} \tag{13}$$

and subsequently extract the spatial location u_l of the maximum activation within it

$$u_l = \underset{(h,w)\in\Omega}{\operatorname{argmax}} \, H_{l:o_l}(h, w) \tag{14}$$

where $\Omega = \{1, \dots, H\} \times \{1, \dots, W\}$. The localization distance is computed as Euclidean distance between predicted location u_l and the ground-truth part location g_l. Averaging over all L parts yields the mean localization distance

$$\mathcal{D} = \frac{1}{L} \sum_{l=1}^{L} \|u_l - g_l\|_2. \tag{15}$$

Furthermore, we quantify *area-based localization* utilizing ground-truth part segmentations of [3]. Analogously to the location-based metric, we first extract the heatmap $H_{l:o_l}$ with the maximum concept score $a_{l:o_l}$ for each group $l \in \{1, \dots, L\}$. Each heatmap is then binarized by thresholding at its 0.5-quantile, retaining the top 50% of pixels by activation. Finally, we compute the Intersection over Union (IoU) [17] with the ground-truth segmentation mask of the corresponding part. Averaging over all parts yields the mean IoU (mIoU). Combined, these two metrics enable us to effectively evaluate the concept localization of models.

Implementation. We closely follow the implementation of [11, 27] to enable a fair comparison. The concept mapping is learned by fine-tuning an Inception-v3 backbone [25], initialized with ImageNet-pretrained weights [6], and localizing prototypes in its feature maps. The learnable prototype vectors are initialized by sampling from a uniform distribution in range $[0, 2 \cdot 10^{-4})$ following [27] and optimized jointly with the model parameters. The class mapping is performed using a single linear layer. ProtoCBM and LocoCBM are trained on the train split of CUB for 1000 epochs with a batch size of 64 and a constant learning rate of $\eta = 0.01$. During training, we apply augmentations consisting of color jitter, random resized cropping to 299×299 and random horizontal flipping. Images are subsequently normalized with $\mu_{\text{norm}} = 0.5$ and $\sigma_{\text{norm}} = 2$. At inference time, the data augmentations are replaced with a center crop operation to the same image size. We utilize the SGD optimizer [20] with momentum $\mu_{\text{SGD}} = 0.9$ and weight decay $\lambda_{\text{SGD}} = 4 \cdot 10^{-5}$. The loss weights were obtained through hyperparameter search on the validation set over 100 trials (*cf*. Tab. 1). The binary cross-entropy loss on concept predictions $\mathcal{L}_{\text{CO}}$ is weighted by the ratios of class imbalances of each concept, following [11]. Our models are implemented using PyTorch. All experiments are conducted on a single NVIDIA L40S GPU with 48 GB of memory, with an average runtime of 15.3 ± 0.2s per training epoch. Early stopping is employed if neither class nor concept validation accuracy improved in 100 epochs.

4.2. Concept-Based Image Classification

To assess its effectiveness, we train ProtoCBM and LocoCBM on CUB in all modes on the task of concept-based image classification. In Tab. 2, we compare to CBMs [11].

Task and concept accuracy. In terms of concept accuracy $\mathcal{A}_{\text{con}}$, our models perform on-par with CBMs across all modes, indicating that explicit localization of concept prototypes enables reliable prediction of concept presences. Investigating the class accuracy $\mathcal{A}_{\text{cls}}$, we outperform CBMs in sequential training mode ($+0.6\%$), demonstrating that concept scores inferred from prototype localization allow for accurate class predictions. In joint mode, however, ProtoCBM shows slightly degraded concept accuracy (-1.0%) and noticeably reduced class accuracy (-5.8%) compared to [11]. We attribute this to the joint objective, as it allows the classification loss to directly influence the concept mapping, which may move the model away from learning semantically meaningful prototypes. Notably, no substantial differences

Mode	λ_{CO}	λ_{CP}	λ_{DC}	λ_{DL}	Criterion
Sequential	1	0.0038	$2 \cdot 10^{-5}$	–	$\mathcal{A}_{con}$
Independent	1	0.0038	$2 \cdot 10^{-5}$	–	$\mathcal{A}_{con}$
Joint	7.8	0.001	0.001	–	$\mathcal{A}_{con} + \mathcal{A}_{cls}$
LocoCBM (all)	4.9	0.0018	0.0009	0.877	$\mathcal{A}_{con} + \mathcal{A}_{cls}$

Table 1. **Loss weights obtained via hyperparameter search.** For each mode, the loss weight corresponds to the configuration that maximized the respective selection criterion over 100 trials.

Model	Accuracy		Localization	
	$\mathcal{A}_{cls}$ (%, ↑)	$\mathcal{A}_{con}$ (%, ↑)	$\mathcal{D}$ (px, ↓)	mIoU (%, ↑)
Sequential				
CBM [11]	75.7±0.3	**96.6±0.1**	98.7±3.3	5.5±0.1
ProtoCBM (ours)	**76.3±0.3**	**96.6±0.0**	**71.4±0.2**	**5.8±0.1**
LocoCBM (ours)	75.2±0.3	96.3±0.0	36.8±1.4	7.05±0.0
Independent				
CBM [11]	**76.0±0.6**	**96.6±0.1**	98.7±3.3	5.5±0.1
ProtoCBM (ours)	74.4±0.8	**96.6±0.0**	**71.4±0.2**	**5.8±0.1**
LocoCBM (ours)	72.6±1.0	96.3±0.0	36.8±1.4	7.05±0.0
Joint				
CBM [11]	**80.1±0.3**	**96.9±0.0**	117.6±24.9	5.4±0.1
ProtoCBM (ours)	74.3±0.6	95.9±0.0	**72.4±0.0**	**6.2±0.1**
LocoCBM (ours)	75.4±0.6	96.1±0.2	39.9±0.7	7.07±0.0

Table 2. **Comparison to CBMs.** We compare our approach with CBMs [11] in image classification on CUB [26] across all modes w.r.t. class accuracy ($\mathcal{A}_{cls}$), concept accuracy ($\mathcal{A}_{con}$), localization distance ($\mathcal{D}$) and mIoU. We report mean ± std. over three seeds. Accuracies for CBMs are taken from [11] and localization metrics computed using official weights. Best localization-unsupervised results per mode are in **bold**. ProtoCBM and LocoCBM perform on-par with CBMs while substantially improving localization.

w.r.t. task and concept accuracy can be observed between ProtoCBM and LocoCBM, indicating that explicit localization supervision does not impede performance.

Concept localization. While prior work primarily evaluates CBMs with respect to concept and class accuracy [11], these metrics are not sufficient to capture whether concept predictions are grounded in the correct image regions. As demonstrated by [13, 19], CBMs fail to properly localize concepts, often attending to unrelated input features when predicting concept scores. Thus, we additionally compare our approach to CBMs with respect to concept localization in Tab. 2. In both localization metrics, ProtoCBM achieves substantially better results across all modes, *e.g.* 71.4px in $\mathcal{D}$ and 5.8% mIoU *vs.* 98.7px in $\mathcal{D}$ and 5.5% mIoU of CBMs in sequential and independent mode. These results indicate that our model more reliably attends to the correct locations when assessing concept presences. This suggests that the integration of learnable prototypes into CBMs and their subsequent localization helps the model to more accurately ground predictions in properly localized concepts. Notably, despite worse task performance of ProtoCBM in joint mode, we maintain superior concept localization (72.4px of ProtoCBM *vs.* 117.6px of CBMs in $\mathcal{D}$). In contrast, the localization of CBMs further degrades in a joint objective ($\mathcal{D}$ of 117.2px) compared to sequential and independent modes ($\mathcal{D}$ of 98.7px), despite higher concept and class accuracy. This further supports the hypothesis that classification supervision on the concept mapping is suboptimal, as it can encourage the model to attend to incorrect spatial locations and exploit spurious correlations, thus rendering the model less interpretable. By integrating explicit supervision on concept locations, LocoCBM reduces $\mathcal{D}$ by 54% compared to ProtoCBM and further improves upon mIoU by +1.3%. Overall, the improved concept localization of our approach, combined with its strong concept and class accuracy, indicates that our model is more faithful in concept-based image classification compared to CBMs. By grounding its concept predictions in more accurately localized concepts, our approach adheres to the goals of CBMs more closely.

Qualitative examples. Looking at the qualitative examples presented in Fig. 3, we see how ProtoCBM visibly improves concept localization and how this translates to more accurate predictions of concept scores. In multiple instances, CBMs attend to unrelated spatial locations, hence the model fails to correctly predict the presence of a concept (*e.g. col. 2*, *col. 5*). In contrast, ProtoCBM more accurately localizes concepts, which enables reliable assessments of concept scores (*e.g.* the bill in *col. 1*, the nape in *col. 6*). Moreover, the CBM heatmaps allude to reliance on spurious correlations. In some instances, CBMs attend to unrelated image regions yet correctly predict concept presences (*e.g. col. 4*, *col. 6*), indicating that predictions are not grounded in visual evidence. This is mitigated by ProtoCBM, as its activations are more compactly and consistently centered on the correct concept locations, whereas CBM activations are more scattered. Overall, the qualitative results further demonstrate that concept prototype learning improves the spatial grounding of CBMs, therefore enabling more faithful interpretability.

4.3. Generalization and Robustness

In this section, we conduct multiple experiments to investigate whether the improved concept localization of ProtoCBM and LocoCBM translates into improved generalization and robustness under distribution shifts. We evaluate both CBMs and our approach across three settings:

Generalization under background shifts. First, we assess generalization under background shifts by evaluating CUB-trained models on Waterbirds [21]. The results in Tab. 3 *(a)* demonstrate that ProtoCBM generalizes better compared to CBMs: While the classification accuracy of CBMs degrade substantially (75.7% → 48.4%), our model remains more robust (76.3% → 62.9%), therefore adapting better to background shifts. At the same time, our model slightly improves

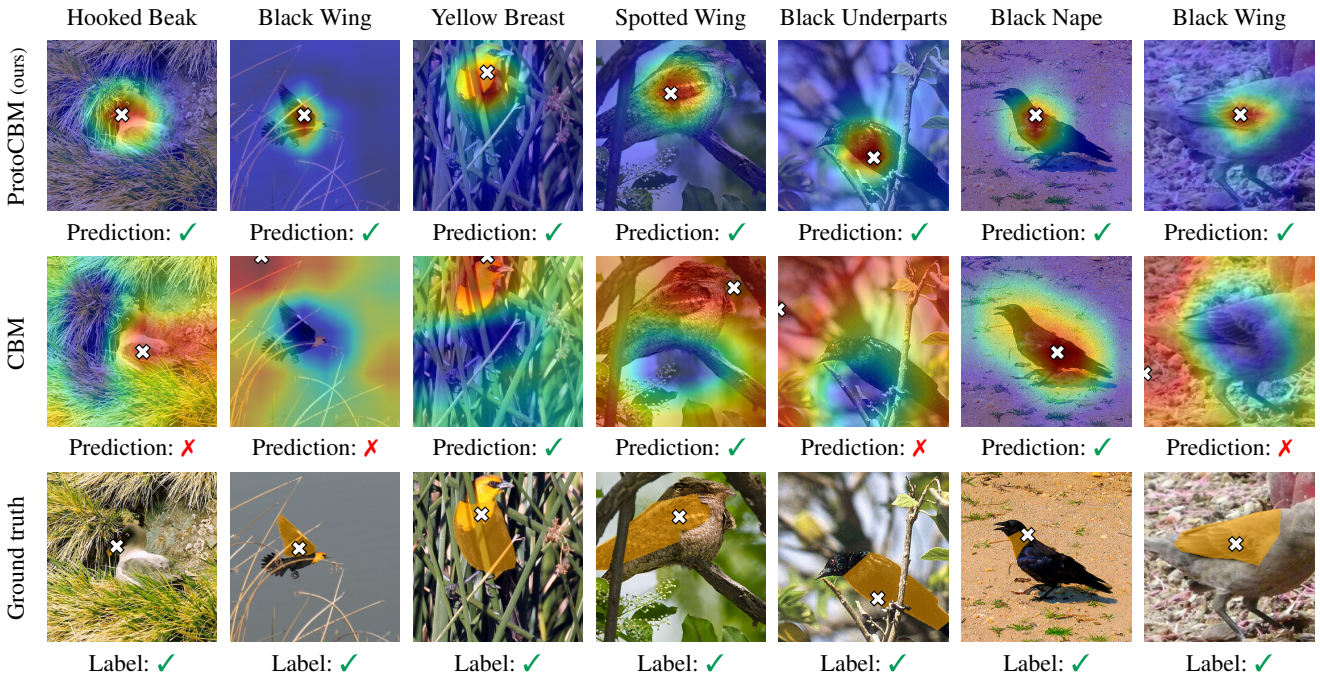


Figure 3. **Visualizations of concept localization.** Exemplary concept activation maps of ProtoCBM and CBMs [11], indicating the activation of a given concept at each location (red = higher). The cross marks the location of maximal activation. Below each image we report whether the concept is predicted as present (✓) or absent (✗). Ground-truth concept locations [26], segmentations [3] and labels are provided in the last row. ProtoCBM grounds its concept predictions in the correct regions, whereas CBMs attend to irrelevant locations.

concept localization (-0.8px in $\mathcal{D}$), whereas the localization of CBMs further deteriorate ($+2$px in $\mathcal{D}$). This suggests that the explicit localization of concept prototypes can enable improved generalization, as learned concept prototypes can be re-used in novel contexts, thereby reducing the reliance on dataset-specific information. This is further supported by the superior performance of LocoCBM ($+1.2\%$ in $\mathcal{A}_{\text{cls}}$ and -28.7px in $\mathcal{D}$), as the supervision on concept locations enables further improvements in generalization capacities.

Robustness to spurious correlations. Next, we examine the robustness to spurious background correlations using Waterbirds [21]. In its train split, spurious correlations are artificially introduced by shifting the co-occurrences of birds to background scenes: Most water birds are set to appear in water scenes and most land birds are depicted on land. At test time, this co-occurrence is balanced so that both bird types are equally likely to appear in either scene. As shown in Tab. 3 *(b)*, ProtoCBM maintains similar concept accuracy while substantially outperforming CBMs in task accuracy (70.8%), where CBMs exhibit degraded results (59.9%). These findings indicate that grounding concept predictions through accurately localized prototypes mitigates reliance on irrelevant background features that can encode spurious correlations. This also reflects in the increase of $+0.3\%$ in task performance of LocoCBM compared to ProtoCBM. The degraded concept localization of CBMs further supports this

Model	Accuracy		Localization	
	$\mathcal{A}_{\text{cls}}$ (%, ↑)	$\mathcal{A}_{\text{con}}$ (%, ↑)	$\mathcal{D}$ (px, ↓)	mIoU (%, ↑)
***(a)* CUB → Waterbirds**				
CBM [11]	48.4±2.0	93.8±0.2	100.7±3.1	5.4 ±0.1
ProtoCBM (ours)	**62.9±0.5**	**93.9±0.2**	**70.6±0.2**	**5.7±0.0**
LocoCBM (ours)	64.1±1.2	94.0±0.1	41.9±2.5	7.0±0.0
***(b)* Waterbirds (unbalanced) → Waterbirds (balanced)**				
CBM [11]	59.9±0.7	**95.8±0.1**	118.6±0.9	4.9±0.0
ProtoCBM (ours)	**70.8±0.3**	95.7±0.0	**71.2±0.4**	**5.8±0.1**
LocoCBM (ours)	71.1±1.0	95.4±0.1	39.0±1.5	7.06±0.0

Table 3. **Generalization and robustness to background shifts.** Comparison of ProtoCBM, LocoCBM and CBMs [11] (sequential) in terms of *(a)* generalization by training on CUB [26] and evaluation on Waterbirds [21], and *(b)* robustness to background shifts by training on Waterbirds with biased backgrounds and testing on a set with a balanced background distribution. We report mean ± std. over three seeds. Our models adapt better in both settings.

($\mathcal{D}$ of 118.6px compared to 98.7px in CUB-trained CBMs), as it indicates that CBMs are more likely attend to irrelevant regions, resulting in the impeded task performance.

Concept substitutions. Finally, we evaluate the robustness to distribution shifts via concept substitutions using SUB [2]. In this benchmark, synthetic bird images are generated in which a target concept is substituted with another concept value atypical for its species (*e.g.* the *blue crown* of a *blue*

jay is replaced with a red one). We measure whether CUB-trained models correctly detect the inserted concept (denoted as $\mathcal{S}^+$) and whether the removed concept is predicted as absent ($\mathcal{S}^-$). In Tab. 4, we report the performance of CBMs and our models. ProtoCBM more reliably detects substituted concepts ($\mathcal{S}^+$ of 37.8% in sequential mode) compared to CBMs ($\mathcal{S}^+$ of 35.1%), demonstrating better adaption to local concept changes. However, at the same time, it also detects removed attributes more frequently ($\mathcal{S}^-$ of 50.4%) than CBMs ($\mathcal{S}^-$ of 52.0%), alluding to a tradeoff between the ability to detect new concepts and suppressing old concept co-occurrences. Through further localization supervision, LocoCBM mitigates this issue, as it reliably improves upon both metrics ($\mathcal{S}^+$ of 41.9% and $\mathcal{S}^-$ of 52.3%).

Overall, these experiments demonstrate that improving concept localization in CBMs can enhance their generalization and robustness under distribution shifts. By accurately grounding concepts in images, ProtoCBM generalizes better under background shifts and is more robust to spurious correlations and concept substitutions. Thus, the integration of explicit localization of concept prototypes can help to make CBMs more faithful and interpretable in image classification.

4.4. Influence of Loss Terms

As proposed by [27], we introduced additional loss terms compared to CBMs that aim to improve prototype learning. To measure their impact, we conduct an ablation study in which we start with solely the classification loss $\mathcal{L}_{\text{CL}}$ and gradually include further loss terms. We report the results after training each combination on CUB [26] in Tab. 5. Each term contributes meaningful information during training of ProtoCBM, as each further improves the accuracy and localization of our model. The class-level supervision alone is not sufficient to enable accurate prediction and localization of concepts. Integrating concept-level supervision $\mathcal{L}_{\text{CO}}$ induces a substantial increase in both concept accuracy ($+44.7\%$) and localization (-19.8px in $\mathcal{D}$). However, solely adding $\mathcal{L}_{\text{CO}}$ does not reliably improve the task performance. Both the compactness criterion $\mathcal{L}_{\text{CP}}$ and the decorrelation loss $\mathcal{L}_{\text{DC}}$ counteract this, as they lead to better classification results ($\mathcal{L}_{\text{CP}}$ induces $+0.5\%$ and $\mathcal{L}_{\text{DC}}$ $+0.7\%$ $\mathcal{A}_{\text{cls}}$) while further marginally improving the concept accuracy. Overall, these results indicate that concept supervision coupled with prototype learning regularization is necessary to jointly achieve good task performance and reliable concept localization.

5. Conclusions and Limitations

We introduce ProtoCBM, a method to improve the concept localization in Concept Bottleneck Models (CBMs) [11] by learning prototypes that localize concepts within images. We demonstrate that our approach achieves on-par concept and class accuracy while improving both proposed localization metrics. Extensive analysis shows that ProtoCBM more faithfully attends to relevant image regions compared to CBMs. By grounding its predictions in accurately localized concepts, ProtoCBM exhibits greater robustness under distribution shifts and better generalization to unseen compositions. In addition, we propose LocoCBM, where we impose additional supervision on concept locations, further improving its faithfulness. Our findings imply that improved task performance and increased visual interpretability are not a tradeoff, but can rather be achieved simultaneously. Overall, we show that integrating prototype learning for concept prediction into CBMs is a promising direction for future research on improving model interpretability.

Limitations and future work. Despite its improved concept localization, the activation maps of ProtoCBM often focus solely on the center of the bird instead of specific parts, which $\mathcal{L}_{\text{DC}}$ only partially mitigates. This issue may arise from the low-resolution CNN feature maps of 8×8, but additional experiments with different backbones are required to verify this. Additionally, our investigation is limited to concept-level interpretability. As seen in Tab. 3, CBMs and ProtoCBM exhibit a discrepancy of 14% in task performance. This may be due to information leakage [8] or differing active concepts provided to the classifier. Therefore, additional research on the interaction between the intermediate bottleneck and the classifier may be required to further deepen the understanding and interpretability of CBMs. Including prototype learning into state-of-the-art CBM variants is a promising direction for future work that can further validate its effectiveness.

Mode	Model	$\mathcal{S}^+$ (%, ↑)	$\mathcal{S}^-$ (%, ↑)
Sequential / Independent	CBM [11]	35.1±3.1	**52.0±2.3**
	ProtoCBM (ours)	**37.8±2.7**	50.4±1.6
	LocoCBM (ours)	41.9±1.4	52.3±3.0
Joint	CBM [11]	34.3±1.1	**56.6±2.3**
	ProtoCBM (ours)	**38.9±0.7**	48.7±1.8
	LocoCBM (ours)	43.7±2.7	52.1±1.0

Table 4. **Robustness to concept substitutions.** We evaluate CBM [11], LocoCBM and ProtoCBM on SUB [2] to assess robustness to concept distribution shifts (mean ± std. over three seeds). Reported metrics (all in %) indicate how often the substituted attribute ($\mathcal{S}^+$) and not the removed attribute ($\mathcal{S}^-$) are detected. Our approach adapts better to concept substitutions.

Losses				Accuracy		Localization	
$\mathcal{L}_{\text{CL}}$	$\mathcal{L}_{\text{CO}}$	$\mathcal{L}_{\text{CP}}$	$\mathcal{L}_{\text{DC}}$	$\mathcal{A}_{\text{cls}}$ (%, ↑)	$\mathcal{A}_{\text{con}}$ (%, ↑)	$\mathcal{D}$ (px, ↓)	mIoU (%, ↑)
✓				73.2±1.2	50.9±1.1	92.2±1.7	6.1±0.2
✓	✓			73.2±0.8	95.6±0.1	**72.4±0.5**	6.1±0.1
✓	✓	✓		73.7±0.7	95.8±0.2	**72.4±0.4**	6.1±0.1
✓	✓	✓	✓	**74.2±0.6**	**95.9±0.0**	**72.4±0.4**	**6.2±0.1**

Table 5. **Influence of loss terms.** We ablate the influence of loss terms by subsequently adding them to ProtoCBM (Joint) and training each combination on CUB [26]. We report mean ± std. over three seeds. Each term improves the accuracy and localization.

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ProtoCBM: Improving Localization of Concept Bottleneck Models with Prototype Learning

Supplementary Material

A. GitHub Project Info

The GitHub repository can be found here. Note that we originally started working in a fork of the CBM [11] codebase before transferring our results to this repository. The GitHub IDs of all group members are:

- MarcSaghir
- mel-oezkan
- pascal0012
- bock-sebastian

B. Individual Contributions

Below, we list the key contributions of each group member. However, it should be noted that for most contributions, all team members were engaged up to some extent, *e.g.* in discussions or debugging:

Marc Saghir

- Midterm presentation
- Implementation / Reproduction of paper [18] for investigation of CBM locality (deprioritized and discontinued)
- Reproduction of APNs [27]
- Implementation and evaluation of original bounding box metric for localization [27, 31]
- Implementation of visualizations
- Implementation and evaluation of adapted distance based localization metric
- Implementation of dataset structure
- Final report (primarily abstract, introduction and related works)

Melih Özkan

- Midterm presentation
- Reproduction and implementation of CBMs
- Implementation of supervision on concept localization into our model (LocoCBM)
- Implementation and evaluation of our model and CBM on the SUB benchmark [2]
- Investigation into the reliability of the SUB dataset
- Investigation into the correlation / relationship between localization and task and concept accuracies (not reported in paper due to space limitations)
- Investigation into the distribution of concept predictions and subsequent comparison to CBMs
- Final report

Pascal Makossa

- Pitch presentation
- Initial implementation of Joint prototype learning in the original CBM codebase
- Added option for multiple prototypes per concept (discarded due to poor results)
- Hyperparameter tuning and Experiment scripts for extensive evaluation
- Rewrite to modular codebase for ProtoCBM
- Implementation of Sequential & Independent modes for ProtoCBM
- Added loss term to reduce intro-group activation map variance (discarded due to poor results)
- Hyperparameter tuning for different ProtoCBM setups & multiple objectives
- Final report (primarily method section & Sec. D)

Sebastian Bock

- Pitch presentation
- Implementation of the first version of ProtoCBM in the original CBM codebase
- Implementation of the *area-based* localization metric
- Investigation into the reliability of the localization metric proposed by [31]
- Reproduction of APNs [27]
- Rewrite to modular codebase for ProtoCBM
- Implementation of visualizations
- Implementation and investigation into the robustness of models using Waterbirds [21]
- Ablation w.r.t. impact of each loss term on performance
- Running the final experiments to gather results as reported in paper
- Reproduction and evaluation of CBM across all of our benchmarks
- Final report (primarily experiments section)

C. Compute infrastructure

All our computations were performed on the ADA cluster after some initial delay to get access. A single NVIDIA L40S GPU was more than sufficient for end-to-end training of our model including finetuning of the pretrained CNN backbone on the CUB dataset [26].

We were sporadically restricted by limited availability of the cluster throughout the semester and occasional queue backlog, as well as some inexplicable termination of longer-running processes (*e.g.* hyperparameter tuning).

D. Concept Visualizations

Considering a central aspect of our evaluation is how well the model localizes concepts, we want to investigate the most relevant image locations for concept prediction during inference. Since ProtoCBM already computes concept activations at each location in the feature map of the backbone, we only need to normalize them to $[0, 1]$ using the maximum a_i^+ and minimum a_i^- activations to obtain concept visualizations

$$H_i^P(x_{h,w}) = \frac{A_i(x_{h,w}) - a_i^-(x)}{a_i^+(x) - a_i^-(x) + \varepsilon}. \tag{16}$$

We add a constant $\varepsilon = 10^{-7}$ for numerical stability. Due to the lack of explicit activation maps in vanilla CBMs, we make use of Gradient-Weighted Class Activation Maps (Grad-CAM) [22]. Concretely, for a given sample x, we compute the gradient δ_{ij} of the i-th concept score w.r.t the j-th feature in the CNN outputs of the last layer $b(x)[j]$:

$$\delta_{ij} = \frac{\partial}{\partial b(x)[j]} m(b(x))_i \tag{17}$$

Using these scores as weights, we obtain the activation map $H_i^C(x)$ for concept i as a linear combination of feature maps

$$H_i^C(x) = \sum_j \delta_{ij} b(x)[j] \tag{18}$$

which we subsequently normalize similar to H^P. Finally, we upsample the concept maps of CBM $\left(H^C\right)$ and our model $\left(H^P\right)$ to the original image size $H \times W$.

E. Discussion on the Reliability of Localization Metrics

Zhang et al. [31] introduce an area-based localization metric for concepts on the CUB dataset, which is further used in subsequent literature [27, 28]. This metric utilizes the bird bounding boxes and the point locations of the bird body parts provided by [26]. First, synthetic ground-truth bounding boxes are constructed for each body part by drawing a box around each corresponding part location. The box side length is chosen to be equal to a quarter of the length of the bird bounding box. If none is provided, a quarter of the image height or width is chosen. The same box length is used for each body part. During inference, it then constructs bounding boxes around predicted locations in the same fashion and calculates the IoU [17] between the paired ground truth and prediction boxes. We argue that this method is flawed, in that it presumes a fixed, arbitrary bounding box size for body parts with varying sizes. Utilizing area based metrics, such as bounding boxes, is not necessarily meaningful if the area is independent of the actual part and the bird size. Drawing arbitrary bounding boxes does not encourage compact segmentation predictions that are relative to the size of the ground truth mask. Additionally, it may falsely imply a correct localization. For example, predicting the location of a small body part of a large bird results in two large bounding boxes to be compared. Therefore even if the part is completely missed, the boxes may show some overlap. Conversely, correctly localizing a non-convex body part may result in slightly misplaced bounding boxes, which may show less overlap than appropriate.

F. Implementation Details

This section provides an overview of the categorization of CUB-200 [26] and CUB-70 [3] concepts into part related groups, as well as information regarding training, validation and test splits of our benchmarks.

CUB-200. CUB concepts have intrinsic subgroups, *e.g.* different throat colors can be all categorized into the subgroup of throat color. These subgroups can again be summarized into part groups, which is useful for the part-wise concept prototype mapping we adopt from [27] in our approach. For convenience and comparability, we adopt their group definitions, which we present in Tab. 6.

Group	Subgroups
Belly	Belly Pattern, Belly Color
Breast	Breast Pattern, Breast Color, Throat Color
Back	Back Color, Back Pattern
Head	Bill Shape, Bill Color, Bill Length, Forehead Color, Nape Color, Eye Color, Head Pattern, Crown Color
Tail	Upper Tail Color, Under Tail Color, Tail Pattern, Tail Shape
Leg	Leg Color
Wing	Wing Color, Wing Pattern, Wing Shape
Others	Bird Size, Bird Shape, Primary Color, Upper Parts Color, Under Parts Color

Table 6. **Eight part groups of Xu et al. [27] for CUB concepts.** Individual concepts are grouped into subgroups, *e.g.* all variations of wing colors. These subgroups are then classified into the part groups seen in the table.

CUB-70. Behzadi-Khormouji and Oramas [3] provide segmentation masks for eleven part groups, which are different than the eight in Tab. 6. To evaluate on this dataset, we matched the part location groups of the CUB [26] dataset to the provided part segmentation groups. This allows us to map between attributes associated with CUB parts and the corresponding segmentation masks.

Dataset Splits. For each dataset, we utilize the splits provided by their respective official repository. We provide links

Segmentation Group	CUB Groups
Beak	Beak
Body	Back, Belly, Breast, Throat
Head	Crown, Forehead
Neck	Nape
Eye	Left Eye, Right Eye
Leg	Left Leg, Right Leg
Wing	Left Wing, Right Wing
Tail	Tail

Table 7. **CUB-70 part segmentation groups and corresponding CUB part groups.** The eleven segmentation groups get reduced to eight, by combining groups with left and right variations, namely eyes, wings and legs.

for each dataset in the following: CUB [26], CUB-70 [3], SUB [2] and Waterbirds [21]